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Analytics Skills for Business

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①  
a) Git is the version control system to create collaborative analytics projects. They possess a repository (local) that allows for working ~~offline~~ <sup>online</sup>. \* Meanwhile GitHub was created to work with teams and has an online repository. In Git every member of the team can work on the project simultaneously and GitHub allows it to be shared. It is also important to notice that GitHub allows to pull scripts as well.

\* the word  
is offline.

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b) Repository: Is where the projects are stored and for collaborative work it saves all the versions of the project and it allows transparency and the possibility to audit

Commit: Is a snapshot of the project at a certain point in time. Is important as it allows to review all the project

Branch: Is an individual working line. It allows to collaborative work and each team member to work in their own branch.

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②  
a) Python allows to process data, make graphical ~~rep~~ representations but also it is useful for general programming so it is widely used among data scientist. Meanwhile SQL is useful to graphic the data, and transform it.

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⑥ Libraries: It is a group of data previously made. With these data you can work on applying functions, graphs, etc. It is important as allows to show data without entering manual data.

Variables: They are the elements that have an effect on what we are programming. They are important because without them there will be nothing to program or visualize.

Conditions: They are ~~the same as loops~~ "if". For example something can happen if other thing is ~~the~~ true or false. They provide complexity to the analysis.

⑦ Docker can store images ~~and~~ that ~~can~~ can be accessed.

⑧ IDE are development environments where you can make scripts and handle data easily. ~~the~~ You can develop complex projects and visualize them.

- R studio
- VS Code.

m/zt



③ Tableau: it is a widely used tool in data analytics to create visual dashboards. Transforming raw data into visually friendly information.

Power BI: It is made by <sup>Microsoft</sup> ~~unibers~~ and is similar to Tableau, it allows to create dashboards from ~~the~~ different data. For example you can ~~easy~~ easily take an excel data and transform it into a dashboard.

Snowflake: it can create conditional scenarios. Is useful to <sup>map</sup> ~~map~~ process and ensure they are correct.

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④ One advantage of using no-code/low-code tools is that they tend to be more user friendly, at the same time it doesn't need deep knowledge in coding languages to successfully use them. Using these tools might be preferred when it is ~~intend~~ intended to use it internally with data that is not too complicated to process or straight forward.

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④ Mean: it is the average. Which means it is the sum of all the values divided between the number of values. It is most appropriate to use it when there are no outliers.

Median: It is the value in the middle when the data is arranged. It can be used when there are outliers as a way to locate the real center of the data.

Mode: It is the most common number of the data. It can be used when there is presence of outliers.

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⑤ Univariate: ~~multiple~~ a single variable in the data, for example the sales of one month.

bivariate: 2 variables in the data. For example <sup>the</sup> yearly sales report containing each month and sales.

multivariate: multiple variables. For example a <sup>yearly</sup> sales report containing sales, month, area, etc.

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② Missin values: when some data is incomplete for example in ~~some~~ ~~columns~~ columns, some have data, some have not. The strategies would be deletion, or ~~the~~ replacement of the data

Inconsistent data: when for example the dates have different formats. Or some words are in captions and others are not. (New York, NEW YORK). The strategie ~~would~~ would be to make ~~the~~ the data uniform via ~~coding~~ ~~coding~~ coding. Also Flaging works as well

Duplicated records: The same data is showed multiple times, creating ~~confusion~~ confusion. By coding ~~first~~ first is to identify where the duplicated records are and then to delete them.

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⑤

(a)  $\text{result} \leftarrow 3 + \text{sqrt}(9) - 2 \wedge \{2\} * 4/3$

(b) `getwd()`

(c) `? mean()`

(d) `library("tidyverse")`

`region <- c("North", "South", "East")`

`sales <- c(4500, 3800, 5200)`

`sales_data <- tibble(region, sales)`

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